

Research paper

A Chiral SU(2) Gauge-Sector Extension and the Quaia $z \sim 1-1.5$ Angular-Mode Diagnostics

Technical regeneration with formulas, notation, data provenance, program inventory, and July 14-16, 2026 residual-global, spatial-block, and injection-calibration claim gates

Field	Value
Prepared for	Clive Boyd
Regenerated	July 16, 2026
Primary empirical focus	Quaia object-level redshift dipole diagnostics around $z = 0.95-1.45$
Theory focus	Phenomenological chiral SU(2)-like gauge-sector effective dark-energy component
Current claim status	Exploratory targeted follow-up; the angular mode is not spatial-null-scale robust and is not an SU(2) detection.

Claim Boundary

The current evidence supports a disciplined follow-up programme around a targeted Quaia redshift window. After scan-law and catalogue-colour regression, the object-level residual global scan reaches the Monte Carlo floor for z_{1p00_1p50} at $|b| > 10$ and $|b| > 15$, but the 50,000-draw spatial-block audit raises the targeted-family p-values to 0.01352, 0.09230, and 0.23260 as cell scale increases from 8 to 16 degrees.

It does not establish a detected SU(2) angular symmetry. The historical raw all-window look-elsewhere gate remains ordinary, while the newer residual object-shuffle gate is interesting but not spatial-null-scale robust.

The latest July 16 audit identifies catalogue colour/selection space and Gaia scan-law geometry as mechanism caveats, while also showing that a very strict Delta BIC gate over-rejects injected angular modes at the observed amplitude scale.

Abstract

This paper formalises a two-level research programme. At the theoretical level, it describes how a chiral,

SU(2)-like non-Abelian gauge sector could be parameterised as an effective late-time component in the Friedmann equation without claiming that ordinary Standard Model SU(2)_L is itself dark energy.

At the empirical level, it documents the July 14-16, 2026 diagnostics on Quaia angular redshift modes, residual-global object shuffles, spatial-block nulls, raw-MU supernova calibration, HHT residual screening, and SPARC rotation-curve posterior checks.

The most interesting empirical hook remains a targeted Quaia mode in the $z = 1.00$ -1.50 family. After regressing redshift on seven Gaia scan-law and catalogue-colour controls, the object-level residual global scan gives residual SNR = 5.7536 and global $p = 0.000999$ for $z1p00_1p50$, $|b| > 10$, with a nearly identical $|b| > 15$ row.

However, the historical raw all-window look-elsewhere test remains ordinary, and the new residual object-shuffle result is not the final spatial null. The 50,000-draw targeted-family spatial wild bootstrap is scale-sensitive: $p_{\text{block}} = 0.01352$, 0.09230, and 0.23260 for 8, 12, and 16 degree cells. Catalogue-quality and scan-law caveats therefore remain stronger than a physical interpretation.

The correct conclusion is therefore not discovery, but a preregistered external-validation gate. An independent catalogue, a fixed angular statistic, a documented mask, and a spatial null chosen before inspection must reproduce the direction and amplitude class before the mode can be promoted.

Executive Claim Boundary

Gate	Observed status	Readout
Theoretical basis	SU(2) gauge sectors are mathematically serious and self-interacting.	Model-building support only; no dark-energy claim follows from Yang-Mills alone.
Targeted z window	broad 0.95-1.45 joint $p = 4.99975\text{e-}5$; narrow 1.151.35 joint $p = 0.00444978$.	Interesting targeted follow-up signal.
Global and spatial nulls	Raw historical all-window $p_{\text{global}} = 0.432713$. Scan-law/colour residual object-shuffle global scan reaches $p = 0.000999$ for $z1p00_1p50$ at $ b > 10$ and $ b > 15$. Shared spatial-block wild bootstrap gives targeted-family $p = 0.01352$, 0.09230, and 0.23260 at 8, 12, and 16 degrees.	Residual object-shuffle signal is a targeted candidate only; spatial-block scale sensitivity blocks promotion.
Angular/footprint proxies	Amplitude ratios about 1.02; direction separations below 5 deg.	These simple angular and selection proxies do not explain the mode.
External maps	SFD, WISE depth, Gaia stellar density retain amplitude ratio about 0.98-1.10.	Dust/depth/stellar-density controls do not absorb the mode.

Gaia scan law	Scan-law preserving null joint $p = 0.00349825$; regression can reduce amplitude, and the colour-template mechanism gate finds Gaia scan-law as the leading carrier for Gaia BP-RP and W1-W2 angular imbalance.	Rare under object-level residual shuffles, but scan-law geometry remains a leading mechanism caveat and the spatial-block null prevents promotion.
Catalogue quality	Matched all-catalogue-quality gate: amp ratio = 0.3097, direction rotation = 121.39 deg, SNR max = 1.876, matched-null joint $p = 0.986$.	Promotion gate not passed; catalogue colour/selection is currently the leading explanation.
Multipoles	Quadrupole-only readout stable; all-controls-plus-quadrupole has q/d mean = 19.1919.	Low-ell structure must be explained before a single-dipole story.
Adjacent streams	Raw-MU is calibration-budget first; SPARC robust split convergence is not passed; HHT remains screening.	Contextual diagnostics, not independent SU(2) detections.
Colour mechanism	Gaia BP-RP and W1-W2 imbalance is best absorbed by Gaia scan-law; SFD dust is only the strongest simple BP-RP correlate.	Survey-geometry caveat; external validation required.

Acronyms and Variables

Acronym	Meaning
AIC	Akaike information criterion; lower values indicate better penalized fit.
BIC	Bayesian information criterion; in this project often used as a BIC-like score with random-effect priors.
CMB	Cosmic microwave background.
CPL	Chevallier-Polarski-Linder equation-of-state form $w(a) = w_0 + w_a(1-a)$.
DES-Dovekie	Recalibrated Dark Energy Survey supernova release used in raw-MU diagnostics.
FITS	Flexible Image Transport System, used for Quaia catalogue, randoms, and maps.
HHT	Hilbert-Huang transform, used here for residual-mode screening.
IMF	Intrinsic mode function in HHT decomposition.
IRSA	NASA/IPAC Infrared Science Archive.
ΛCDM	Lambda cold dark matter cosmological model.
PLAMB	Project shorthand for the logarithmic/optical-depth modified rotation or expansion diagnostics.
Quaia	Gaia-unWISE all-sky quasar catalogue used for angular redshift-mode tests.
RAR	Radial acceleration relation baseline used in SPARC rotation-curve comparison.
SFD	Schlegel-Finkbeiner-Davis dust map.
SN	Type Ia supernova in this document.
SNR	Signal-to-noise ratio; for dipoles, $\sqrt{d^T \text{Cov}(d)^{-1} d}$.
SPARC	Spitzer Photometry and Accurate Rotation Curves database.
SU(2)	Special unitary group of degree 2, the non-Abelian internal symmetry group considered here.
WISE	Wide-field Infrared Survey Explorer; unWISE/AllWISE information enters Quaia and depth controls.
Symbol	Meaning in this paper

$z, z_{\text{HEL}}, z_{\text{CMB}}, z_{\text{HD}}$	Redshift, heliocentric redshift, CMB-frame redshift, and Hubble-diagram redshift with external-flow correction.
μ	Distance modulus, usually $\mu = 5 \log_{10}(D_{\text{L}}/\text{Mpc}) + 25$.
H_0	Present expansion rate in km/s/Mpc.
$E(z)$	Dimensionless expansion function $H(z)/H_0$.
$\Omega_m, \Omega_r, \Omega_k, \Omega_\Lambda$	Matter, radiation, curvature, and cosmological-constant density fractions.
Ω_{chi}	Effective fractional density assigned to the proposed chiral SU(2)-like component.
w_0, w_{chi}	CPL equation-of-state parameters for the effective chiral component.
A_μ^a	SU(2) gauge potential; μ is spacetime index, a is internal group index.
$F_{\mu\nu}^a$	Non-Abelian SU(2) field strength tensor.
g	Gauge coupling in field-theory formulas; also a Gaia G-band magnitude variable in catalogue-quality tables.
Φ	Symmetry-breaking scalar/order parameter in the schematic field-theory extension.
λ_{chi}	Coefficient of a parity-odd chiral term ΦF_{tilde} .
$\mathbf{n}_i$	Unit sky vector for source i in Galactic Cartesian coordinates.
$\mathbf{d}$	Three-component fitted dipole vector in the Quaia redshift regression.
T_{ki}	Template/control value k evaluated at object i .
b_{cut}	Minimum absolute Galactic latitude cut, $ b > b_{\text{cut}}$.

Historical and Theoretical Background

Yang-Mills local SU(2) symmetry

Yang and Mills introduced the modern local non-Abelian gauge principle: if an internal isotopic spin rotation is allowed to vary from point to point, an ordinary derivative no longer transforms covariantly.

A gauge connection must be introduced, and its field strength contains a commutator/self-interaction term. This is the mathematical reason an SU(2)-like sector is a serious candidate for a dynamical cosmological component; it carries stress-energy and can self-interact.

The same fact is also why the claim must be cautious: the existence of a non-Abelian gauge field does not itself imply a late-time darkenergy sector.

$$\begin{aligned} \text{SU}(2) &= \{ U \text{ in } C^{(2 \times 2)} : U^\dagger U = I, \det(U) = 1 \} \\ T^a &= \sigma^a / 2, \\ [T^a, T^b] &= i \epsilon^{abc} T^c \quad \psi(x) \rightarrow S(x) \psi(x), \\ D_\mu &= \partial_\mu - i g A_\mu^a T^a \\ F_{\mu\nu}^a &= \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g \epsilon^{abc} A_\mu^b A_\nu^c \end{aligned}$$

Chirality and electroweak precedent

The Standard Model already contains a chiral SU(2) structure: SU(2)_L × U(1)_Y breaks to U(1)_{EM}, with left-handed fermions transforming differently from right-handed fermions.

This motivates a handedness-biased extension but does not identify ordinary weak-interaction vacuum energy with dark energy.

The electroweak scale is far above the observed late-time dark-energy scale.

The viable hypothesis is instead a residual, hidden, emergent, or effective SU(2)-like chiral sector whose parameters can be tested against cosmological data.

$$\text{SU}(2)_L \times \text{U}(1)_Y \rightarrow \text{U}(1)_{\text{EM}}$$

Field-theory testbed

A schematic gravitationally coupled chiral gauge-sector action can be written as a Yang-Mills term, an order-parameter kinetic term, a potential, matter, and an optional parity-odd chiral term.

The parity-odd term is a natural location for a handedness bias, but it must ultimately be constrained with parity-sensitive observables such as CMB TB/EB, cosmic birefringence, or chiral gravitational waves.

$$\begin{aligned}
S = \int d^4x \sqrt{-g} \big[& R/(16 \pi G) - (1/4) F_{\mu\nu}^a F_a^{\mu\nu} \\
& - (1/2) (D_\mu \Phi)^a (D^\mu \Phi)^a - V(\Phi) \\
& + \lambda_{\chi} \Phi F_{\mu\nu}^a \tilde{F}_a^{\mu\nu} + L_m \big] \\
T_{\mu\nu}(A) = & F_{\mu\alpha}^a F_{\nu}^{\alpha a} - (1/4) g_{\mu\nu} F_{\alpha\beta}^a F^{\alpha\beta a}
\end{aligned}$$

Isotropic cosmic-triad ansatz

A single vector field normally selects a preferred spatial direction.

An SU(2) field has a useful cosmological route: lock the internal group index to the spatial index.

The cosmic-triad ansatz preserves isotropy at the background level while retaining non-Abelian dynamics.

$$\begin{aligned}
A_i^a(t) &= \phi(t) \delta_{i^a}, \\
A_0^a &= 0 \\
ds^2 &= -dt^2 + a(t)^2 \delta_{ij} dx^i dx^j
\end{aligned}$$

Phenomenological expansion-history branch

The first-stage project implementation remains phenomenological.

Rather than solving the full SU(2) field equations, the current fitting branch adds an effective density $\Omega_{\chi}(z)$ to the Friedmann function and tests whether supernova, BAO, and CMB-distance data permit it under identical cuts and priors used for LCDM, KdS, FR, and PLAMB-style branches.

$$\begin{aligned}
E^2(z) &= \Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_k(1+z)^2 + \Omega_L + \Omega_{\chi}(z) \\
w_{\chi}(z) &= w_0 + w_a z/(1+z) \\
\Omega_{\chi}(z) &= \Omega_{\chi 0} (1+z)^{3(1+w_0+w_a)} \exp[-3 w_a z/(1+z)]
\end{aligned}$$

Statistical Methods

Quaia redshift dipole regression

The Quaia angular diagnostic fits the object-level Quaia redshift as a function of sky direction and optional standardised templates.

For each redshift window and Galactic latitude cut, each object receives a unit vector n_i in Galactic Cartesian coordinates.

The fitted vector d is interpreted only as an empirical angular mode, not as a physical velocity or field direction.

```
z_i      = c0 + d . n_i + sum_k gamma_k standardize(T_{k,i}) + epsilon_i
A_dipole = ||d||
SNR_d    = sqrt( d^T Cov(d)^(-1) d )
direction(d) = (l, b) from d / ||d||
```

Direction stability is summarised by pairwise angular separation across latitude cuts.

A low separation means the fitted direction remains coherent when the mask changes.

Template stability is summarised by the amplitude ratio relative to the no-template baseline and by rotation of the fitted direction.

Null p-values and look-elsewhere correction

```
p_upper = (1 + count(metric_mock >= metric_observed)) / (1 + N_mock)
p_lower = (1 + count(metric_mock <= metric_observed)) / (1 + N_mock)
p_global = probability that any scanned window/cut reaches the observed
threshold
```

The targeted stability p-values use pre-selected windows after the direction-amplitude scan.

They measure whether a selected follow-up target is unusual under nested random-sky mocks.

The global look-elsewhere p-values scan a larger family and are the appropriate gate for a headline anisotropy claim.

In this project, the targeted and residual object-shuffle gates are interesting, but the spatial-block gate is the stricter promotion test and is not passed across plausible cell scales.

Raw-MU hierarchical offset likelihood

```
r = MU_observed - MU_model(H0, beta)
r = X_dataset delta_dataset + X_survey delta_survey +
noise delta_dataset ~ Normal(0, sigma_dataset^2)
delta_survey ~ Normal(0, sigma_survey^2)
```

The July 14 supernova result promotes calibration-aware raw-MU offsets as the default diagnostic.

This prevents an apparent exact-log or beta preference from being read as cosmology before dataset and survey calibration structure has been absorbed with explicit priors.

Information criteria

```
AIC      = chi^2 + 2k
BIC      = chi^2 + k ln(N)
Delta AIC = AIC_model - AIC_LCDM
```

Datasets Used

Dataset	Local/project artifact	Purpose
Quaia G<20 quasar catalogue	Cosmology/quaia_G20.0.fits	Primary object-level redshift and sky-position source for angular-mode tests.
Quaia random catalogue	Cosmology/quaia_G20p0_randoms_simple.fits	Random-sky and footprint comparison support.
Quaia selection map	.spyder-py3/selection_function_NSIDE64_G20.0.fits	Selection/depth proxy template.
SFD dust maps	external_datasets/sfd and sfd_dataverse_manifest_2026-07-15.json	External Galactic dust template gate.
Gaia nominal scan law	gaiascanlaw data/cache when available	Scan count and scan angle controls; scan-law preserving redshift shuffles.
AllWISE W1/W2 coverage	external_datasets/allwise_atlas_inventory/allwise_p3am_cdd_w1w2_coverage.csv	Tile-depth control for WISE-related selection effects.
Gaia EDR3 density HiPS	external_datasets/gaia_dr3_density and gaia_edr3_density_hips	External stellar-density control.
Pantheon+SH0ES	PantheonPlusSH0ES/DataRelease; local raw-MU extracts	SN raw-MU calibration and redshift-frame audit.
DES-Dovekie	des-science/DES-SN5YR; local HD tables	SN raw-MU calibration and redshift-frame audit.
Union3.1 / UNITY	Union3p1_UNITY1p8 compressed rows	Compressed SN node included in hierarchical rawMU joint likelihood.
SPARC rotation curves	SPARC local sample and point CSVs; filtered all-Q2 subsets	RAR/PLAMB posterior and stress tests.
HHT residual series	SN residual exports, BAO rows, chronometer rows	Residual-mode and surrogate screening.

Program Inventory

Program or script	Role in this paper
analyze_su2_quaia_locked_templates_multipoles_2026-07-15.py	Locked primary/control redshift windows; template and dipole-plus-quadrupole diagnostics.
analyze_su2_quaia_quality_strata_2026-07-15.py	Quartile splits by zerr, Gaia G/BP-RP, WISE W1/W2/colour, and proper-motion error variables.
analyze_su2_quaia_quality_preserving_nulls_2026-07-15.py	Quality-stratified random-sky and quality-footprint redshift-shuffle nulls.
analyze_su2_quaia_footprint_shuffle_highstat_202607-15.py	High-stat 5,000 mock footprint-preserving redshift shuffles.
analyze_su2_quaia_external_dust_gate_2026-07-15.py	SFD dust, Quaia selection/depth, Gaia scan-law, and local catalogue-quality template regressions.
analyze_su2_quaia_scanlaw_followup_2026-07-15.py	Scan-law preserving shuffles and residualization after scan-law controls.
analyze_su2_quaia_wise_stellar_gate_2026-07-15.py	AllWISE W1/W2 tile-depth and Gaia EDR3 stellar-density external template gates.
fetch_sfd_dataverse_2026-07-15.py	SFD dust acquisition and manifest capture.
fetch_gaia_dr3_density_hpx5_2026-07-15.py / fetch_gaia_dr3_density_uws_2026-07-15.py	Gaia density acquisition attempts and HiPS-backed fallback.
run_su2_quaia_overnight_highstat_pipeline.ps1	Overnight high-stat targeted stability and global look-elsewhere pipeline.
fit_sparc_hierarchical_map.py / analyze_sparc_posterior_run.py	SPARC MAP/posterior stress scoring and convergence readouts.
rawmu calibration-budget scripts	Hierarchical dataset/survey offset likelihood and zHD/zHEL/zCMB frame policy.
HHT residual export and resonance scripts	Historical residual screening, IMF summaries, and surrogate controls.
analyze_su2_quaia_matched_quality_gate_2026-07-15.py	Propensity-overlap matched-quality gate for zerr, Gaia colour, WISE colour, PM errors, and full catalogue-quality blocks.
analyze_su2_quaia_colour_decomposition_gate_2026-07-15.py	Decomposes catalogue-quality suppression into optical colour, infrared colour, redshift-uncertainty interactions, and compact colour-redshift estimator blocks.
analyze_su2_quaia_colour_template_mechanism_2026-07-15.py	Tests whether SFD, selection/depth, Gaia scan-law, WISE depth, and stellar-density templates carry the colour-field dipole/hemisphere imbalance.
analyze_su2_quaia_spatial_block_collinearity_2026-07-16.py	Runs common-sign equal-area spatial wild bootstraps across nested latitude cuts and quantifies locked-direction template collinearity.
audit_sparc_posterior_diagnostics_2026-07-16.py	Computes rank-normalised split Rhat, approximate bulk ESS, MCSE and paired posterior in-sample fit-density checks.

Research History Breakdown

Period/date	Milestone	Consequence for claim status
1954	Yang-Mills local isotopic gauge invariance establishes non-Abelian SU(2) self-interaction.	Mathematical foundation for gauge-sector models, not direct evidence for dark energy.
Electroweak era	SU(2) _L x U(1) _Y supplies a chiral Standard Model precedent.	Handedness is theoretically natural, but ordinary weak SU(2) is not the late-time component.
Pre-July 2026	Effective SU2_chiral_DE branch proposed with Omega_chi0, w0_chi, wa_chi, and dormant chi_bias.	Sets up a testable phenomenological branch.
July 14, 2026	Quaia direction-amplitude audit finds the stronger stable window at 0.95-1.45 and narrower target around 1.15-1.35.	Narrows the candidate angular target; broad symmetry wording rejected.
July 14, 2026	High-stat redshift-window mock catalogues generated before interpreting anisotropy scans.	Prevents premature interpretation of redshift-window scans.
July 14, 2026	Raw-MU likelihood promoted to hierarchical dataset/survey offsets; zHD preregistered as default frame.	Supernova exact-log preferences become calibration/frame diagnostics.
July 14, 2026	SPARC RAR rescue reaches classic max Rhat = 1.031; robust split audit pending.	Classic convergence improved, but this milestone is superseded by the 16 July rank-split audit.
July 14, 2026	HHT/CMB-SN archives captured and summarised.	Useful screening method; not a confirmed coupling signal.
July 15, 2026	Overnight targeted and global Quaia mocks complete.	Targeted signal survives; global all-window look-elsewhere does not.
July 15, 2026	Locked template, quality strata, scan-law, WISE-depth, and stellar density gates run.	External maps mostly do not absorb signal; local quality controls remain blocking caveat.
July 15, 2026 evening	Matched catalogue-quality gate and colour-decomposition gate complete.	All-catalogue-quality and colour-redshift estimator blocks fail promotion gates; colour/selection caveat becomes stronger than the earlier simple-template readout.
July 15, 2026 evening	Colour-template mechanism gate complete.	Gaia scan-law is the leading carrier for BP-RP and W1-W2 angular imbalance; SFD dust remains the strongest simple BP-RP correlation but not the main angular mechanism.
July 16, 2026	50,000-draw spatial-block null and direction-template collinearity audit.	Targeted-family significance becomes spatial-scale sensitive; discovery language is further restricted.
July 16, 2026	Rank-split SPARC convergence and paired-nuisance diagnostic audit.	RAR rescue is partial, PLAMB remains non-converged, and earlier stress scores are relabelled as prior-redrawn rather than posterior-predictive.

July 14-15 Quaia Results

Direction-amplitude perturbation

Window	Mean amplitude	Max SNR	Max separation	Readout
0.90-1.40	0.0023115	3.6887	17.33 deg	stable direction
0.95-1.45	0.0030202	4.7852	12.00 deg	strongest and stable
1.00-1.50	0.0023450	3.7059	17.94 deg	baseline stable
1.05-1.55	0.0013951	2.7582	61.34 deg	direction breaks
1.10-1.60	0.0010442	2.1157	76.01 deg	direction breaks

The scan does not support wording such as a stable broad SU(2) angular symmetry.

It supports a targeted follow-up near $z = 0.95-1.45$, with a narrower 1.15-1.35 control window.

Upward perturbations rotate the direction and weaken the amplitude.

Targeted stability vs global look-elsewhere

Target	Observed max SNR	Direction spread	p(SNR)	p(spread)	Joint p
broad 0.95-1.45	4.7852	13.4723 deg	0.000149993	0.0040998	0.0000499975
narrow 1.15-1.35	3.29694	18.5137 deg	0.0380481	0.0170991	0.00444978

Residual-global readout: after scan-law/colour regression, the object-level all-window correction reaches $p = 0.000999$ for the z_{1p00_1p50} , $|b| > 10$ residual SNR and priority.

Target	bcut	Observed SNR	Global p	Priority p	$z_{1p00_family\ p}$
z_{1p00_1p50}	35	3.70594	0.432713	0.20076	0.0203959
z_{1p00_1p50}	10	3.47955	0.70086	0.394521	0.0407918
z_{1p00_1p50}	15	3.59778	0.563887	0.291542	0.0271946
z_{1p00_1p50}	25	3.49857	0.677864	0.372326	0.0365927

Interpretation

The targeted p-values justify further work on a locked follow-up target.

They do not override the global p-values.

The residual-global object-shuffle audit is a strong follow-up trigger, but the shared spatial-block null is the current headline promotion gate and it is not passed robustly.

Locked template and multipole controls

Model	Amp ratio mean	Amp ratio min	Max direction sep	SNR max	Readout
angular_systematics_proxy	1.0157	1.00692	4.51692 deg	4.83764	pass proxy gate
footprint_selection	1.02118	1.00522	2.4011 deg	4.88309	pass proxy gate
quadrupole_only	1.0126	1.00645	2.90399 deg	4.83856	dipole comparatively stable
catalog_quality	0.584129	0.295773	117.563 deg	4.78188	fails quality gate
all_controls	0.615275	0.335244	116.713 deg	4.74355	selection-sensitive
all_controls_plus_quadrupole	0.531886	0.308058	119.049 deg	4.10654	low-ell structure important
Multipole model	Amp ratio mean	q/d mean	q SNR max	Median delta BIC	Readout
quadrupole_only	1.0126	0.506064	4.45315	45.6913	dipole comparatively stable
all_controls_plus_quadrupole	0.531886	19.1919	9.67658	-102825	low-ell structure important

Quality strata and quality-preserving nulls

Variable	Q4/Q1 amp ratio	Max direction sep	Quality gate
zerr	1.73952	60.2607 deg	fail
g	2.29744	68.773 deg	fail
bp_rp	0.951058	103.846 deg	fail by rotation
w1	0.778735	53.2999 deg	fail
w2	1.40766	49.8262 deg	fail
w1_w2	0.775818	88.0069 deg	fail
pmra_error	2.35282	88.6125 deg	fail
pmdec_error	1.97269	67.1426 deg	fail

High-stat footprint-preserving redshift shuffle p-values for the locked primary window.

Quality variable	n_mocks	p_snr	p_pair_sep	p_coherence	Joint p
zerr	5000	0.00039992	0.0127974	0.00039992	0.00019996
g	5000	0.00059988	0.0117976	0.00059988	0.00039992
w1	5000	0.0263947	0.0219956	0.00959808	0.00379924
w2	5000	0.00159968	0.0163967	0.00119976	0.00079984
w1_w2	5000	0.009998	0.0585883	0.0079984	0.00339932
pmra_error	5000	0.00059988	0.0115977	0.00019996	0.00019996
pmdec_error	5000	0.00019996	0.0119976	0.00019996	0.00019996

The footprint-preserving shuffles are not nullifying the targeted signal, but the quartile splits show that the fitted mode is entangled with object-quality space.

The most conservative reading is that the quality variables are not simple explanations by themselves, but they are strong enough to block physical promotion.

Matched catalogue-quality gate

The July 15 propensity-overlap gate balanced catalogue-quality variables between the two hemispheres defined by each latitude-cut baseline dipole axis, then refit the locked redshift dipole.

Good post-weight balance was achieved, but the all-catalogue-quality block collapsed the mode rather than preserving it. This is now an explicit promotion-blocking failure.

Gate	Amp ratio	Direction rotation	SNR max	Joint p	Readout
zerr_only	1.01	0.47551 deg	4.814	see p-value table	survives matched gate
gaia_colour_quality	0.62955	10.74 deg	3.129	see p-value table	fails matched gate
wise_colour_quality	0.72452	3.861 deg	3.775	see p-value table	fails matched gate
pm_error_quality	1.028	2.593 deg	4.859	see p-value table	survives matched gate
all_catalogue_quality	0.30971	121.39 deg	1.876	0.986	fails matched gate

Interpretation: zerr and proper-motion errors alone survive the matched-quality gate, but Gaia colour, WISE colour, and the full catalogue-quality block do not. The all-catalogue-quality row has amplitude ratio 0.3097, 121.39 deg direction rotation, SNR max 1.876, and matched-quality joint p = 0.986.

Colour-decomposition and colour-template mechanism gates

The next decomposition split the suppressing catalogue block into optical colour, infrared colour, redshift-uncertainty interactions, and external survey-template mechanisms.

```

c_i      = alpha + T_i beta + eta_i
eta_i    = c_i - alpha - T_i beta
SMD_c    = (mean(c | H=1) - mean(c | H=0)) / sigma_pooled
A_resid  = ||d_eta||

```

Rank	Feature block	Amp ratio	Direction rotation	SNR max	Readout
1	all_catalogue_quality	0.30971	121.39 deg	1.876	fails matched gate
2	colour_redshift_estimator_compact	0.41956	27.52 deg	2.452	fails matched gate
3	gaia_wise_colour_cross	0.50814	18.59 deg	2.788	fails matched gate
4	zerr_x_gaia_wise_colour	0.52728	17.96 deg	2.901	fails matched gate
5	gaia_bp_rp_only	0.62661	10.96 deg	3.112	fails matched gate
6	gaia_g_plus_bp_rp	0.62955	10.74 deg	3.129	fails matched gate
7	zerr_x_gaia_colour	0.63929	10.84 deg	3.168	fails matched gate
8	wise_w1_plus_w2	0.71779	3.188 deg	3.853	fails matched gate

Colour-template mechanism readout

Response field	Best template family	Score	R2	Residual amp/raw	SMD absorption	Readout
gaia_bp_rp_colour	gaia_scanlaw	0.49778	0.0064063	0.54683	0.83592	leading angular carrier
wise_w1_w2_colour	gaia_scanlaw	0.4101	0.00026678	0.57196	0.62123	leading angular carrier
gaia_wise_colour_cross	all_external_templates	0.18821	0.0026753	0.58295	-0.82724	weaker/mixed mechanism

Interpretation: SFD dust has the strongest simple correlation with Gaia BP-RP, but Gaia scan-law best absorbs the dipole/hemisphere imbalance that matters for the locked angular-mode gate. The current claim boundary is therefore survey-geometry mediated, not SU(2) discovery language.

External dust, depth, scan-law, and stellar-density gates

Control group	Amp ratio	SNR max	Max direction sep	Readout
baseline_no_template	1.000000	4.7852	13.4723 deg	reference
sfd_dust_external	1.00644	4.66601	12.9543 deg	SFD dust does not absorb
quaia_selection_depth	1.03478	4.79898	13.5025 deg	selection/depth proxy does not absorb
sfd_plus_selection_depth	1.02617	4.71411	12.9733 deg	combined dust/depth stable
gaia_scanlaw_external	0.783971	3.36404	29.8825 deg	material scan-law caveat
sfd_selection_gaia_external	0.866391	3.60328	31.6722 deg	retains substantial residual
all_available_templates	0.499563	2.92202	48.2136 deg	too mixed; diagnostic only

Scan-law null	nmocks	nscan_bins	p_snr	p_pair	p_coherence	Joint p
scan-law preserving z shuffle	2000	63	0.00949525	0.0809595	0.0069965	0.00349825

Control group	Amp ratio	SNR max	Max direction sep	Readout
wise_depth_external	0.978169	4.75555	15.7795 deg	WISE depth does not absorb
gaia_stellar_density_external	1.09983	5.00263	16.0578 deg	stellar density does not absorb
wise_plus_stellar_external	1.06899	4.92309	17.965 deg	combined WISE/stellar stable
all_external_wise_stellar	0.920458	3.55534	35.1206 deg	substantial residual remains
all_external_plus_local_quality	0.711541	3.66543	36.7889 deg	local quality still matters

Adjacent Empirical Streams

Supernova raw-MU calibration and redshift frame

The July 14 raw-MU update changes the default supernova diagnostic.

The correct default is now the hierarchical dataset+survey offset likelihood in the preregistered HD/zHD frame, with calibration-budget sensitivity shown explicitly.

zHEL and zCMB are retained as controls rather than headline frames.

Reference result	Value
Reference row	HD, dataset+survey, 25 mmag survey / 50 mmag dataset, FR_BETA05_H0fixed675
N	3422
Calibration random effects	27
Data chi2 at MAP	3092.819
Calibration prior chi2	19.154
BIC-like score	3146.032
Beta-free H0-fixed beta	0.518263
No-offset HD beta-free control	beta = 0.466532, BIC-like = 4118.004

SPARC PLAMB/RAR status

The RAR convergence-rescue run improved the classic diagnostic, but the 16 July rank-normalised split audit gives max Rhat = 1.0616 and minimum bulk ESS = 76.3. The preregistered approximately ≤ 1.05 convergence target is therefore not yet passed, and PLAMB remains a subset-win story rather than a full-sample win.

Metric	Value or readout
RAR robust convergence	max rank-normalised split Rhat = 1.0616; minimum bulk ESS = 76.3. Improved, but the ≤ 1.05 gate is not passed.
PLAMB robust convergence	max rank-normalised split Rhat = 1.3082; minimum bulk ESS = 8.11. Material non-convergence.
All-Q2 paired fit-density	+13.602 versus RAR; 78 better / 79 worse galaxies. In-sample and provisional.
Low-acceleration outer points	+0.890 paired fit-density. Not cross-validated evidence.
High inclination	+11.927 paired fit-density. Not cross-validated evidence.
Gas dominated	+4.828 paired fit-density. Not cross-validated evidence.
Claim boundary	Subset wins, not a full-sample win; robust convergence and held-out galaxy prediction remain open gates.

HHT CMB/SN residual screening

The HHT archive is valuable as a residual-screening workflow.

It should not be presented as a confirmed CMB-SN coupling signal until a new preregistered run separates probes, controls survey/window effects, and subsequently passes global/surrogate correction.

HHT readout	Value
SN residual weighted RMS	LCDM 0.15159; SU2 0.15548; SU2R 0.15542
Extended SU2-minus-LCDM IMF energy fractions	IMF1 0.6177; IMF2 0.4290; IMF3 0.2068
Surrogate control	max-energy upper-tail $p = 0.9703$; sum-energy upper-tail $p = 0.9604$
Older IMF2-vs-BAO baseline	$r_{DM} = 0.1006$, $p = 0.0078$; $r_{DH} = -0.1037$, $p = 0.0061$
Claim boundary	Candidate-feature and method-development evidence only.

Interpretation and Next Tests

The current SU(2)/Quaia state is unusually productive because the result is neither dead nor promotable.

Simple dust, WISE tile depth, Gaia stellar density, and the local selection map do not simply erase the targeted $z = 0.95$ -1.45 mode, but matched catalogue-quality and colour-template mechanism gates now identify a stronger survey-geometry caveat.

At the same time, spatial-block scale sensitivity, matched all-catalogue-quality collapse, and Gaia scan-law-linked colour imbalance prevent discovery language.

Next test	Reason	Promotion criterion
Cross-catalogue angular validation	Quaia quality variables may encode catalogue construction or photometric-redshift behaviour.	Same z -window direction and amplitude class appears in an independent quasar/LSS catalogue with a preregistered mask.
Survey-template mechanism likelihood	The colour-template mechanism gate points to Gaia scan-law geometry as the leading angular carrier.	A generative scan-law/selection likelihood reduces the colour imbalance without removing a preregistered cross-catalogue angular mode.
Matched quality/colour rescue	The all-catalogue-quality block collapses the locked mode, and the compact colour-redshift estimator block suppresses it strongly.	Mode remains with amplitude ratio >0.8 and direction rotation <30 deg under preregistered matched-quality controls.
Global null with fixed locked target	The all-window global p -value is ordinary.	A future pre-registered target family gives a globally corrected p small enough for headline language.

Multipole decomposition	All-controls-plus-quadrupole shows low-ell structure.	Coherent dipole/quadrupole pattern is explained by a physical or survey model, not just a marginal dipole.
Theory-to-data bridge	Expansion history alone cannot establish chirality.	A predicted angular/parity observable is derived from the SU(2) sector and beats survey-template alternatives.

Bottom-line conclusion

Use the phrase: targeted Quaia $z \sim 1-1.5$ angular-mode follow-up with catalogue-colour and scan-law gates. Do not use: detected SU(2) angular symmetry, cosmological anisotropy detection, or full proof of a chiral gauge-sector dark-energy component.

July 16 Model-Upgrade Reproducibility Audit

Changing the Codex model affects subsequent reasoning, code generation, and prose only. It does not alter completed chains, random seeds, source files, logs, or Git history. The upgrade was therefore treated as an independent code-and-method audit rather than as a reason to reinterpret numerical output automatically.

Spatial-block family null

Residual-global object-shuffle context

The residual-global scan regressed z against Gaia scan count $\log_{10} p$, scan-angle $\cos^2$ mean, scan-angle $\sin^2$ mean, scan-angle resultant, Gaia BP-RP z -score, WISE W1-W2 z -score, and their cross term for every scan window with $N \geq 50$. In the object-level residual-shuffle null, the z_{1p00_1p50} , $|b| > 10$ row had residual SNR = 5.7536, residual priority = 0.05524, and global $p = 0.000999$ for both SNR and priority.

That result is an important follow-up trigger, not a discovery gate. The same observed family maximum is no longer below 1% once residuals are sign-flipped by shared spatial blocks at all tested cell scales.

The $z = 1.00$ - 1.50 family was re-tested across $|b|$ cuts of 10, 15, 20, 25, 30, and 35 degrees after regression on seven scan-law and colour controls. Equal-area Galactic longitude by $\sin(\text{latitude})$ cells received common Rademacher signs across all nested cuts, preserving cross-cut covariance and coarse angular correlation.

$$\begin{aligned} T_{\text{obs}} &= \max_b \text{SNR}_b \\ r_i^{(m)} &= u_{\{c(i)\}}^{(m)} r_i, \quad u_c^{(m)} \in \{-1, +1\} \\ T_m &= \max_b \text{SNR}_b^{(m)} \\ p_{\text{block}} &= [1 + \sum_m I(T_m \geq T_{\text{obs}})] / (M + 1) \end{aligned}$$

Cell scale	Cells	Mocks	Observed max SNR	Null p99	Family p	Readout
8 deg	910	50,000	5.7536	5.9165	0.01352	Weak tail at finest scale
12 deg	424	50,000	5.7536	7.5140	0.09230	Ordinary
16 deg	231	50,000	5.7536	9.1402	0.23260	Ordinary

Direction-template collinearity

At the locked $|b| > 15$ degree direction, the seven-control projection has $R^2 = 0.16363$ and $VIF = 1.19565$. The largest single correlation is with Gaia scan-angle $\sin(2\theta)$, $r = -0.40438$. The full design condition number is 2.735, so numerical ill-conditioning is not the main problem; moderate physical/template alignment is. The CMB direction is more aligned with the same controls, with $R^2 = 0.41097$.

Direction	Latitude cut	Projection R^2	VIF	Max $ r $	Canonical r	Condition no.
Locked b15	15 deg	0.16363	1.19565	0.40438	0.63785	2.735
CMB	15 deg	0.41097	1.69771	0.64069	0.63785	2.735

Injection and model-selection calibration

The matched injection study recovers approximately half of a one-times injected angular mode on average (mean recovery 0.5012). A strict controlled physical fit retains 0.28335 of that amplitude, while the final

Delta BIC < -10 gate passes only 0.0011 of one-times injections. The BIC threshold is therefore a stringent promotion rule with poor power near the observed scale, not a stand-alone physical rejection test.

SPARC diagnostic correction

The earlier SPARC analyser used classic unsplit Rhat and drew galaxy nuisance parameters from their priors when forming the reported stress scores. Those values are historical prior-redrawn fit checks, not posterior-predictive scores. A paired-draw re-analysis uses each stored global draw with its associated galaxy nuisance draw and adds robust rank-normalised split diagnostics.

Model	Max classic Rhat	Max rank-split Rhat	Min bulk ESS	Gate
RAR	1.03071	1.06163	76.27	Not passed
PLAMB kappa,p	1.05424	1.30819	8.11	Material failure

Claim boundary after audit: SPARC retains interesting subset and MAP patterns, but neither robust convergence nor held-out galaxy prediction supports a full-sample promotion. The defensible wording remains: subset wins, not a full-sample win.

Research-standing priorities

Priority	Action	Promotion gate
1	Independent, preregistered quasar/LSS catalogue validation	Direction and amplitude class reproduced under a fixed mask/statistic and justified spatial null
2	Reparameterise SPARC mass-to-light hierarchy and sample with a proven HMC/NUTS implementation	Rank-split Rhat ≤ 1.01 -1.05, adequate ESS, and stable chain-wise predictive summaries
3	Galaxy-held-out or grouped cross-validation for PLAMB versus RAR	Positive out-of-sample score across all-Q2 and preregistered stress subsets
4	One-command environment, data hashes, licences and CI smoke tests	Fresh clone reproduces key tables without local-path repair

Reproducibility and Archived Artifacts

Artifact group	Representative files
July 15 SU2/Quaia results	github_export/results/2026-07-15/su2_quaia/ and outputs/su2_quaia_*_2026-07-15.*
July 14 SU2/Quaia baseline	github_export/results/2026-07-14/su2_quaia/ and outputs/su2_quaia_*_2026-07-14.*
Code capture	github_export/code/quaia_su2/ plus copied scripts in outputs/
Raw-MU calibration	github_export/results/2026-07-14/rawmu/ and rawmu_calibration_budget_* files
SPARC posterior	github_export/results/2026-07-14/sparc/ and sparc_filtered_rar_rescue_readout_2026-07-14.md
HHT CMB/SN archive	github_export/results/2026-07-14/hht_cmb_sn/ and hht_cmb_sn_scan_readout_2026-07-14.md
Research papers	github_export/docs/papers/ for citation-facing PDFs; local editable DOCX stored in outputs/
Matched-quality and colour mechanism gates	github_export/results/2026-07-15/su2_quaia/su2_quaia_matched_quality_gate_*; su2_quaia_colour_decomposition_gate_*; su2_quaia_colour_template_mechanism_*

References and Source Links

- Yang, C. N., and Mills, R. L. (1954). Conservation of Isotopic Spin and Isotopic Gauge Invariance. *Physical Review* 96, 191195.
- Storey-Fisher, K. et al. (2024). Quaia, the Gaia-unWISE Quasar Catalog: An All-sky Spectroscopic Quasar Sample. *ApJ* 964, 69. arXiv:2306.17749; data at <https://zenodo.org/records/10403370>; code at <https://github.com/kstoreyf/gaia-quasarslss>.
- Quaia local artifacts: C:/Users/clive/Documents/Cosmology/quaia_G20.0.fits; quaia_G20p0_randoms_simple.fits; selection_function_NSIDE64_G20.0.fits. □ SFD dust map source: Harvard Dataverse DOI 10.7910/DVN/EWCNL5, <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/EWCNL5>.
- AllWISE and IRSA programmatic access documentation: https://irsa.ipac.caltech.edu/ibe/ibe_search_api.html and IRSA AllWISE documentation pages.
- Gaia EDR3 density HiPS: <https://alaska.cds.unistra.fr/ancillary/GaiaEDR3/density-map/>.
- gaiascanlaw package and source: <https://pypi.org/project/gaiascanlaw/> and <https://github.com/zpenoyre/gaiascanlaw>.
- Pantheon+SH0ES data release: <https://github.com/PantheonPlusSH0ES/DataRelease>.
- DES-Dovekie / DES-SN5YR repository: <https://github.com/des-science/DES-SN5YR>.
- Union3 / UNITY1.5 reference: Rubin et al., Union Through UNITY, arXiv:2311.12098.
- Project generated reports listed in the reproducibility table, especially su2_quaia_overnight_highstat_status_2026-0715.md, su2_quaia_locked_template_multipole_report_2026-07-15.md, su2_quaia_quality_strata_readout_2026-0715.md, su2_quaia_external_dust_gate_readout_2026-07-15.md, su2_quaia_scanlaw_followup_readout_2026-07-15.md, and su2_quaia_wise_stellar_gate_readout_2026-07-15.md.

Appendix: Acceptance Gates

Gate	Pass threshold used in practice	Current status
Targeted stability	Low targeted p-values for SNR and direction coherence.	Passed for follow-up only.
Global look-elsewhere and spatial null	Small preregistered global p-value that is stable over justified spatial correlation scales.	Not passed; historical all-window $p_{\text{global}} = 0.432713$ and targeted-family spatial-block p ranges from 0.01352 to 0.23260.
Template stability	Amplitude ratio not substantially suppressed; direction rotation below about 30 deg.	Not passed as a promotion gate; SFD/WISE-depth/stellar-density controls are not decisive, but Gaia scan-law and colour-template gates are material caveats.
Quality robustness	Quartile/matched-quality tests do not rotate or amplify/suppress the mode materially.	Not passed; matched all-catalogue-quality gate collapses amplitude to 0.3097 and rotates direction by 121.39 deg.
Multipole coherence	Low-ell pattern supports a coherent physical model.	Open; all-controls quadrupole suggests low-ell complexity.
Adjacent diagnostics	SN, HHT, and SPARC pass their own preregistered gates.	Mixed; raw-MU policy is robust, but SPARC fails robust convergence and HHT remains exploratory screening.
Theory prediction	SU(2) field model predicts the observed angular/parity signal quantitatively.	Not yet implemented.
Colour-template mechanism	External templates do not carry the colour-linked angular imbalance.	Not passed; Gaia scan-law best absorbs BP-RP and W1-W2 hemisphere imbalance.